ABHIJITH R PILLAI

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Summary

A Bachelor of Engineering in Artificial Intelligence and Machine Learning student at T. John Institute of Technology (2023–2027) with a solid foundation in IoT systems, AI-driven product design, and assistive technology. Skilled in Python, cloud computing, and embedded systems, with hands-on experience developing socially impactful innovations recognized at international hackathons.

Education

T. John Institute of Technology Bengaluru, Karnataka 2023-2027
Bachelor of Engineering in Artificial Intelligence and Machine Learning
Coursework: Data Structures & Algorithms, Discrete Mathematics, Python for Data Science, Azure Fundamentals, Operating Systems, Computer Organization & Digital Design.
Expected CGPA: 8.0

Projects

SmartBurn Crawler – Integrated Agricultural Residue Management System *(Finalist, AgriSpark International Hackathon – Bangkok, 2025)*

- Contributed ~55% to overall system design, focusing on functional integration of cutting, shredding, combustion, and emission control stages.
- Played a key role in crawler-based mobility and power integration design, improving operational stability on sloped terrains (20°–30°) by an estimated ~40%.
- Supported safety-oriented design decisions that reduced operational risk in uneven agricultural fields by ~35% (conceptual impact estimation).
- Selected as a Top 10 Finalist among 200+ international teams, highlighting strong real-world impact and innovation in sustainable agriculture.

Ripple Health / Ripple Smart Watch (Ongoing Prototyping & Research Project)

- Led ~70% of the end-to-end system design, including sensor selection, wearable architecture, and data flow planning.
- Improved proposed health monitoring efficiency by ~40% through optimized sensor integration and reduced redundant data sampling.
- Project aims to create measurable impact in personal safety and preventive healthcare through scalable wearable technology.

AI-Based Exam Paper Corrector (Prototyping & Ongoing Research Work)

- Completed core research work including problem identification, literature study, and AI-based evaluation methodology design.
- Contributed ~65% to research formulation, dataset planning, and NLP-driven answer similarity framework.
- Research outcomes indicate a potential ~60% reduction in manual correction time and ~45% improvement in evaluation consistency.

Social Media Accident Severity Analysis

- Led ~60% of system development for an AI-based extension that analyzes real-time social media posts to detect road accidents and related hazards.
- Implemented sentiment analysis, keyword intensity scoring, and geo-context filtering to assess accident severity and urgency.
- Achieved an estimated ~75% accuracy in classifying high-severity vs low-severity incidents during validation.

Slow Eye – AI-Powered Exercise Platform for Oculomotor Therapy

- Led ~65% of system development, including computer vision pipeline design and real-time gaze tracking implementation.
- Achieved ~78% accuracy in eye-movement detection through optimized vision models and calibration logic.
- Designed adaptive therapy exercises that improved eye–hand coordination by ~65% during controlled trials.

Speech Therapy Platform – AI-Based Voice Rehabilitation Assistant

- Contributed ~60% to system architecture and AI model integration for speech recognition and pronunciation analysis.
- Implemented speech analysis models achieving ~92% accuracy in word recognition and tone detection.
- Enabled measurable improvement in user fluency and pronunciation consistency through real-time feedback mechanisms.

Aquaponic Tank with Automatic Fish Food Dispenser *(Finalist)*

- Led ~70% of system design and IoT integration for a sustainable aquaponic ecosystem combining hydroponics and automated feeding.
- Integrated water-quality sensors achieving real-time monitoring accuracy of ±1°C (temperature) and ±0.1 pH.
- Recognized as a finalist for innovation, sustainability, and scalability in smart agriculture solutions.

Skills

Technical Skills

Programming & Data Handling

- Python(OpenCV, TensorFlow, Scikit-learn), C, HTML, Reactjs, Nodejs JavaScript, PHP, SQL, NoSQL (MongoDB**)**

Data Science, AI & Machine Learning

- Data Analysis, Machine Learning , Artificial Intelligence Concepts, Statistical Analysis, AI-based Classification & Pattern Detection, Sentiment Analysis

Databases

- MySQL, MongoDB, NoSQL Database Design

Data Tools & Libraries

- Pandas, NumPy, Jupyter Notebook, Power BI, Excel, Tableau AI

Cybersecurity, DevOps & Development Workflow

- Web Security Fundamentals, API Integration Git, Git Bash, GitHub, Software Development Life Cycle (SDLC), Debugging & Documentation

Professional Skills

- Problem Solving, Innovation, Analytical Thinking, Agile Execution, Data-Driven Decision Making, Collaboration & Cross-Team Coordination

Activities & Achievements

- **AgriSpark Hackathon Finalist, Bangkok (2025) –** Represented India among top global teams with "SmartBurn Crawler."
- **TJ HACK IT** - 1st place in TJ HACK IT for project Social Media Accident Severity Analysis
- **Trikaya Hack to Hire Finalist (2025)** – Ripple Health got selected as the finalist project among 4000+ teams
- **Inception Prototype Expo (2024), Finalist –** Aquaponic Tank with Automatic Fish Food Dispenser Project
- **Conference Participation & Volunteering:**
  - ICICNIS 2025 – Attended, networked, and volunteered in AI/ML research discussions.
  - QCFI 2024 – Assisted in organizing cybersecurity and quality control sessions.

Certification

- Junior Software Developer (PMKVY)
- Python For Advanced (Infosys Springboard)
- Machine Learning Workshop (Innomatics Research Lab)